



Department of Electrical and Electronics Engineering Academic Year 2021 – 2022 (Even Semester)

Degree, Semester & Branch: IV Semester B.E. EEE 'A'

Course Code & Title: IC8451 Control Systems

Name of the Faculty member (s): Mr. S. Meenakshi Sundaravel, AP(SG)/EEE

Innovative Practice Description

- **Unit / Topic:** Unit II / Time response of first order and second order systems
- **Course Outcome:** CO2
- **Topic Learning Outcome:** TLO8
- **Activity Chosen:** MATLAB Simulation
- **Justification:**
MATLAB provides us a huge number of toolboxes to plot graphs, visualize complex datasets. Students use MATLAB to perform complex scientific calculations, it also lets the students to visualize and implement complex mathematical functions. With the help of MATLAB simulation, students can determine the time domain specifications of the given system in an easy way.
- **Time Allotted for the Activity:** 40 minutes
- **Details of the Implementation:**
 - ✓ I have given demonstration on MATLAB simulation of step response of first order and second order systems.
 - ✓ All the students are taken to Control and Instrumentation Lab and the procedure to simulate the given system is explained to them.
 - ✓ Students individually simulated the time response of the given system.

• CO – PO / PSO mapping:

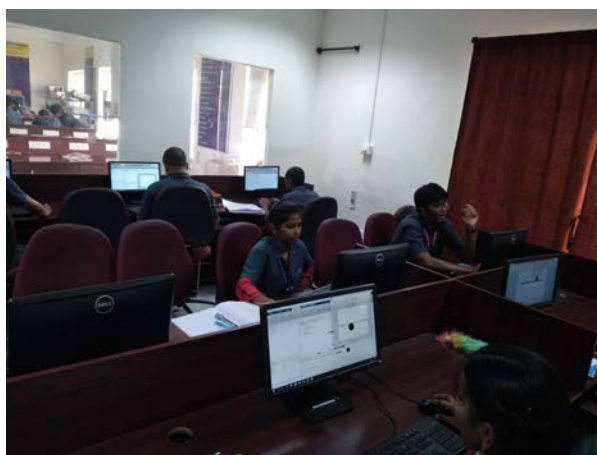
CO	PO1	PO2	PO4	PO5	PSO1
CO2	3	3	2	2	1

• PO / PSO mapped:

Innovative practice	PO1	PO2	PO4	PO5	PSO1
	3	3	2	2	1
Justification for correlation	Student applying their engineering knowledge to understand the concept of time domain	Students apply mathematical concepts to determine the time domain	Students analyze the response of the system with the desired response and	Students use MATLAB simulation software for determining time response of the system and their proficiency in	Student analyze the performance of the given system using MATLAB software Package

	analysis of a given system	specification of the given system	draws necessary conclusion	MATLAB simulation is improved	
--	----------------------------	-----------------------------------	----------------------------	-------------------------------	--

• **Images / Screenshot of the practice:**



• **Reflective Critique:**

❖ ***Feedback of practice from students and other stakeholders:***

Students find this simulation is the easier way to determine the time domain specification of the given system.

Students also assured to continue the simulation using Online MATLAB Simulation

❖ ***Benefit of the practice:***

The time response of a higher order system or any complicated system can be determined using MATLAB simulation

Student understood the time domain specifications clearly and it is reflected during the discussion session.

❖ ***Challenges faced in implementation:***

Few late bloomer students felt difficult in coding and simulating the transfer function

References:

<https://in.mathworks.com/support/learn-with-matlab-tutorials.html>

Signature of Faculty Member

Form No. AC 10c

Rev.No.01

HOD

Effective Date: 02.08.2021



Department of Electrical and Electronics Engineering Academic Year 2021 – 2022 (Even Semester)

Degree, Semester & Branch: IV Semester B.E. EEE 'A'

Course Code & Title: IC8451 Control Systems

Name of the Faculty member (s): Mr. S. Meenakshi Sundaravel, AP(SG)/EEE

Innovative Practice Description

- **Unit / Topic:** Unit III /Bode Plot

- **Course Outcome:** CO 3

- **Topic Learning Outcome:** TLO13

- **Activity Chosen:** MATLAB Simulation

- **Justification:**

MATLAB provides us a huge number of toolboxes to plot graphs, visualize complex datasets. Students use MATLAB to perform complex scientific calculations, it also lets the students to visualize and implement complex mathematical functions. With the help of MATLAB simulation Students can determine the stability of the given system using Bode Plot in an easy way.

- **Time Allotted for the Activity:** 40 minutes

- **Details of the Implementation:**

- ✓ I have given demonstration on MATLAB simulation for obtaining the bode plot of the given system.
- ✓ All the students are taken to Control and Instrumentation Lab and the procedure to simulate the given system is explained to them.
- ✓ Students individually simulated the Bode plot of the given system.

- **CO – PO / PSO mapping:**

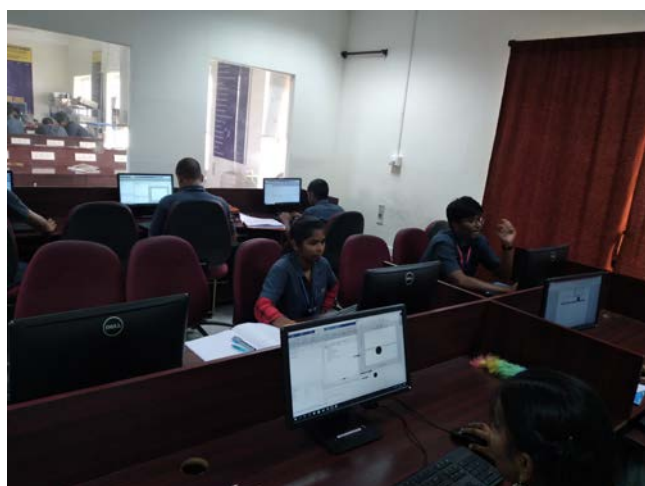
CO	PO1	PO2	PO4	PO5	PSO1
CO2	3	3	2	2	1

- **PO / PSO mapped:**

Innovative practice	PO1	PO2	PO4	PO5	PSO1
	3	3	2	2	1
Justification for correlation	Student applying their engineering knowledge to understand the concept of stability	Students apply mathematical concepts to determine the stability of the given	Students analyze the stability of the system and draws necessary conclusion	Students use MATLAB simulation software for determining time response of the system and their proficiency in	Student analyze the performance of the given system using MATLAB software Package

	analysis of a given system	system		MATLAB simulation is improved	
--	-------------------------------	--------	--	-------------------------------------	--

• **Images / Screenshot of the practice:**



• **Reflective Critique:**

❖ ***Feedback of practice from students and other stakeholders:***

Students find this simulation is the easier way to determine the stability of the given system.

Students also assured to continue the simulation using Online MATLAB Simulation

❖ ***Benefit of the practice:***

The stability of a higher order system or any complicated system can be determined using MATLAB simulation

Student understood the Bode plot concept clearly and it is reflected during the discussion session.

❖ ***Challenges faced in implementation:***

Few late bloomer students felt difficult in coding and simulation

References:

<https://in.mathworks.com/support/learn-with-matlab-tutorials.html>

Signature of Faculty Member

Form No. AC 10c

Rev.No.01

HOD

Effective Date: 02.08.2021



RAMCO INSTITUTE OF TECHNOLOGY

Approved by AICTE, New Delhi & Affiliated to Anna University
Accredited by NAAC & An ISO 9001:2015 Certified Institution
NBA Accredited UG Programs: CSE, EEE, ECE and MECH

Department of Electrical and Electronics Engineering Academic Year 2021 – 2022 (Even Semester)

Degree, Semester & Branch: IV Semester B.E. EEE 'A'

Course Code & Title: IC8451 Control Systems

Name of the Faculty member: Mr. S. Meenakshi Sundaravel, AP(SG)/EEE

Innovative Practice Description

- **Unit / Topic:** Unit I / Basic Elements in Control System – Open and closed loop Systems
- **Course Outcome:** CO 1
- **Topic Learning Outcome:** TLO1
- **Activity Chosen:** Demonstration
- **Justification:**
 - The basic block diagram of control system has various components and the students understand the type of system through various examples. After teaching the concept, I thought of conducting this activity to enhance the learning level of the students and as a teacher I can evaluate the understanding level of the students.

Time Allotted for the Activity: 5 Minutes

- **Details of the Implementation:**

After teaching the concept, students were asked to tell the function of each components of the system.

Total Strength is 31

Reporter: **Myself**

At the end the Class (Last 5 minutes)

- ✓ They can realize the function of basic components of system.
- ✓ I asked the students to think about difference between open and closed loop system.
- ✓ They can able to perform mathematical modeling of the mechanical systems.

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO10	PO12	PSO3
C214.1	3	3	1	1	2

- **PO / PSO mapped:**

PO	Justification
PO1	Students will have fundamental Engineering Knowledge to understand the examples of open and closed loop system
PO2	Students will identify the mathematical, engineering and other relevant knowledge that applies to derive the transfer function of the system.
PO12	Students will be able to submit a clear, well written statement.
PSO3	Students will be able to represent a mechanical system in a block diagram.

- **Images / Screenshot of the practice:**

Reflective Critique:

1. Pre-implementation Reflection:

Benefits:

- Students can able to attend the question even if the questions are in indirect form.
- Students can able to write the concepts in examination without any confusion.

Challenges: Time utilization for conducting activity.

2. Post-implementation Reflection:

Benefits: Students understood the concept which was reflected from their answers during the next class.

Challenges: Late bloomers had some difficulties in answering the questions during the short period of time.

❖ *Benefit of the practice:*

- Most of the students able to recollect the concept quickly whenever asked.
- The success of the activity was evaluated by asking the same question in Internal Assessment test I – Around 80% of students answered.

Signature of Faculty Member

HOD